HCC RECONSTRUCTION

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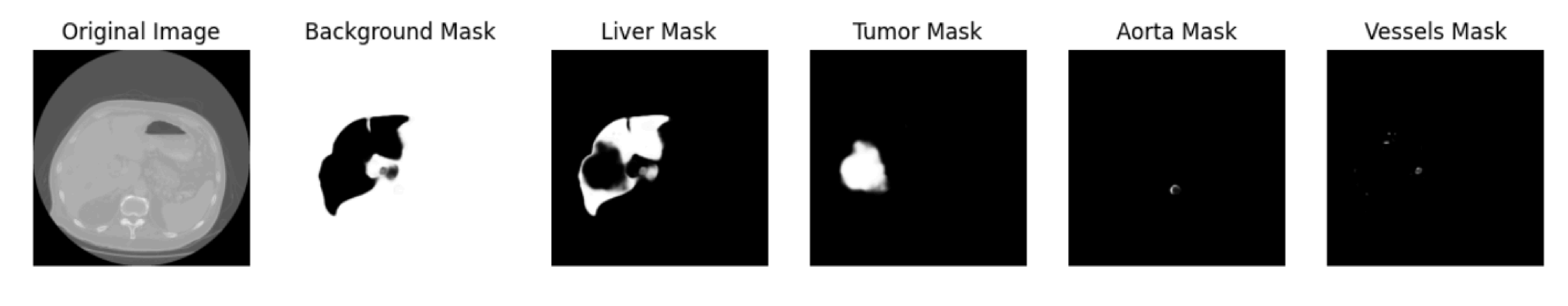
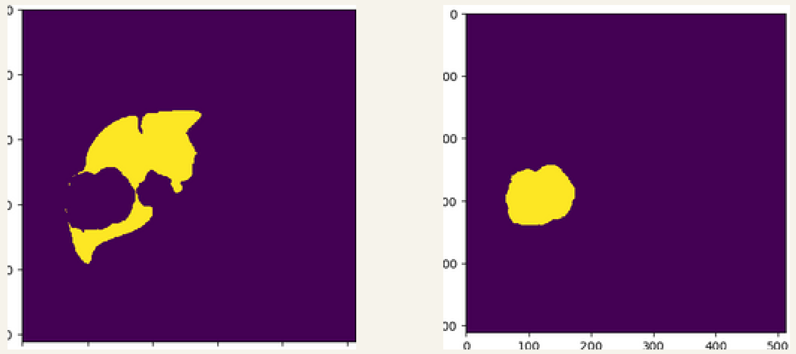
Pre - Trained UNET

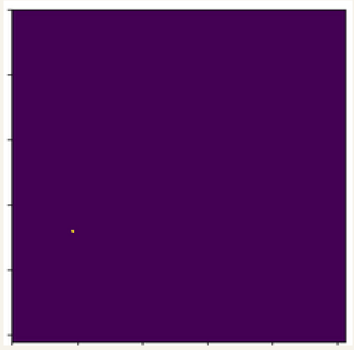
```
Train Dice Scores per Class: {0: 0.9995672697757609, 1: 0.7081782233534755, 2: 0.3387023149594806, 3: 0.1480682400008385, 4: 0.28884091140945434}
Val Dice Scores per Class: {0: 0.9983705311133388, 1: 0.6595943983265693, 2: 0.27354756518595874, 3: 0.1304383363824275, 4: 0.2378602820566794}
Test Dice Scores per Class: {0: 0.9899099744366247, 1: 0.6046695155972602, 2: 0.1252741130908044, 3: 0.0376466498530358, 4: 0.17399381707755732}
```

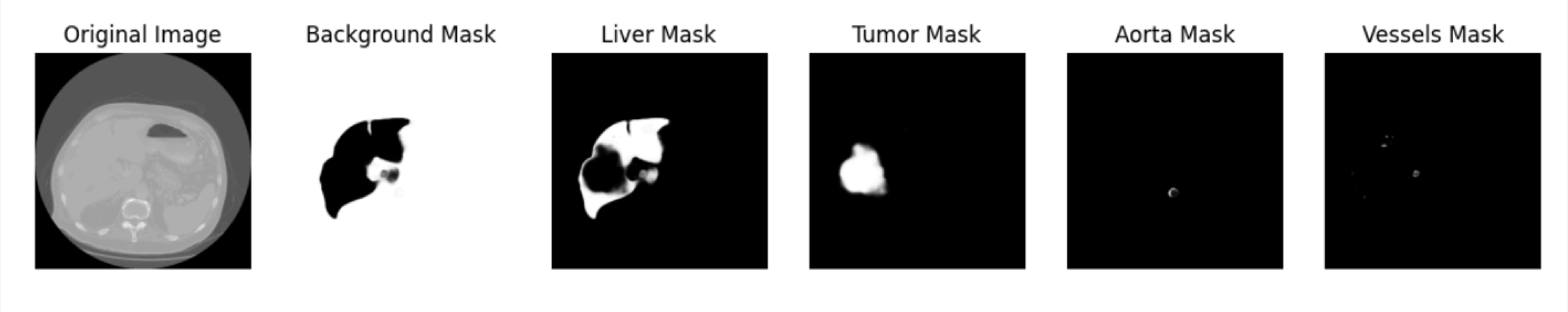


Pre - Trained UNET



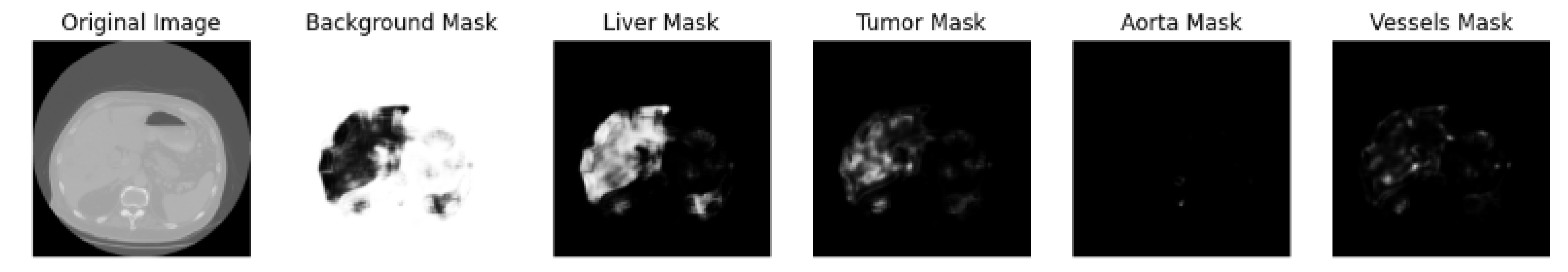
BLANK





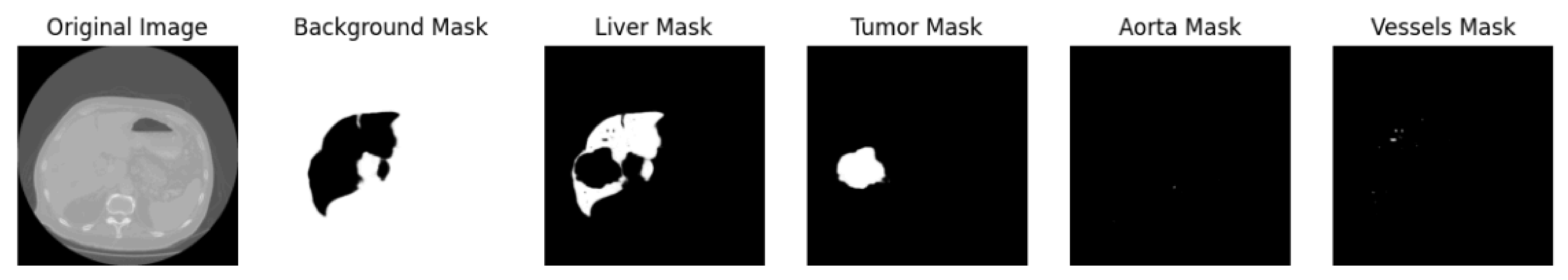
UNET

```
Train Dice Scores per Class: {0: 0.9861519800455185, 1: 0.4075081356968222, 2: 0.054270289
67576307, 3: 0.017120391539006612, 4: 0.03617903823156843}
Val Dice Scores per Class: {0: 0.9863482904836598, 1: 0.41026598199261827, 2: 0.0535294217
4841367, 3: 0.015114381459427659, 4: 0.04184664429116958}
Test Dice Scores per Class: {0: 0.9804280389484757, 1: 0.38902353403226564, 2: 0.034104555
1936299, 3: 0.012103941001410568, 4: 0.03517546080490973 5}
```



ResUNet

```
Train Dice Scores per Class: {0: 0.9988355492459524, 1: 0.6698479262031412, 2: 0.32611127987027, 3:
0.1365106725541798, 4: 0.2739067741213881}
Val Dice Scores per Class: {0: 0.997827875798597, 1: 0.6376999299907755, 2: 0.26799917574637894, 3:
0.12223775057811762, 4: 0.22393746549982718}
Test Dice Scores per Class: {0: 0.9895929306763515, 1: 0.5875347196465853, 2: 0.10148867767223176, 3:
0.04829863654143263, 4: 0.17120336021330965}
```



Pre - Trained UNET this is the best we have got till now, have also tried windowing but that had negative imapct on the results, tried leaky relu and weighted loss functions but there was no improvement

Currently trying to implement BraTS paper - which suggests few improvements over the vanilla UNet, such as Deep supervision.

```
[19]: # Evaluate Dice coefficients
train_dice_scores = evaluate_dice(unet_model, train_dataset)
print("Train Dice Scores per Class:", train_dice_scores)

test_dice_scores = evaluate_dice(unet_model, test_dataset)
print("Val Dice Scores per Class:", test_dice_scores)

true_test_dice_scores = evaluate_dice(unet_model, true_test_dataset)
print("Test Dice Scores per Class:", true_test_dice_scores)
```

```
Train Dice Scores per Class: {0: 0.9986945013743322, 1: 0.6732660947589324, 2: 0.30838
549065996895, 3: 0.13719151507128807, 4: 0.2293046456157265}
Val Dice Scores per Class: {0: 0.9976311056925777, 1: 0.6353606349675956, 2: 0.2549537
8225065934, 3: 0.10685819955333103, 4: 0.21447367566826317}
Test Dice Scores per Class: {0: 0.989335219688853, 1: 0.5793559813184729, 2: 0.1296771
4786479095, 3: 0.04457217054306679, 4: 0.1817909929692418}
```

(2.0 * intersection) / (union + 1e-6)

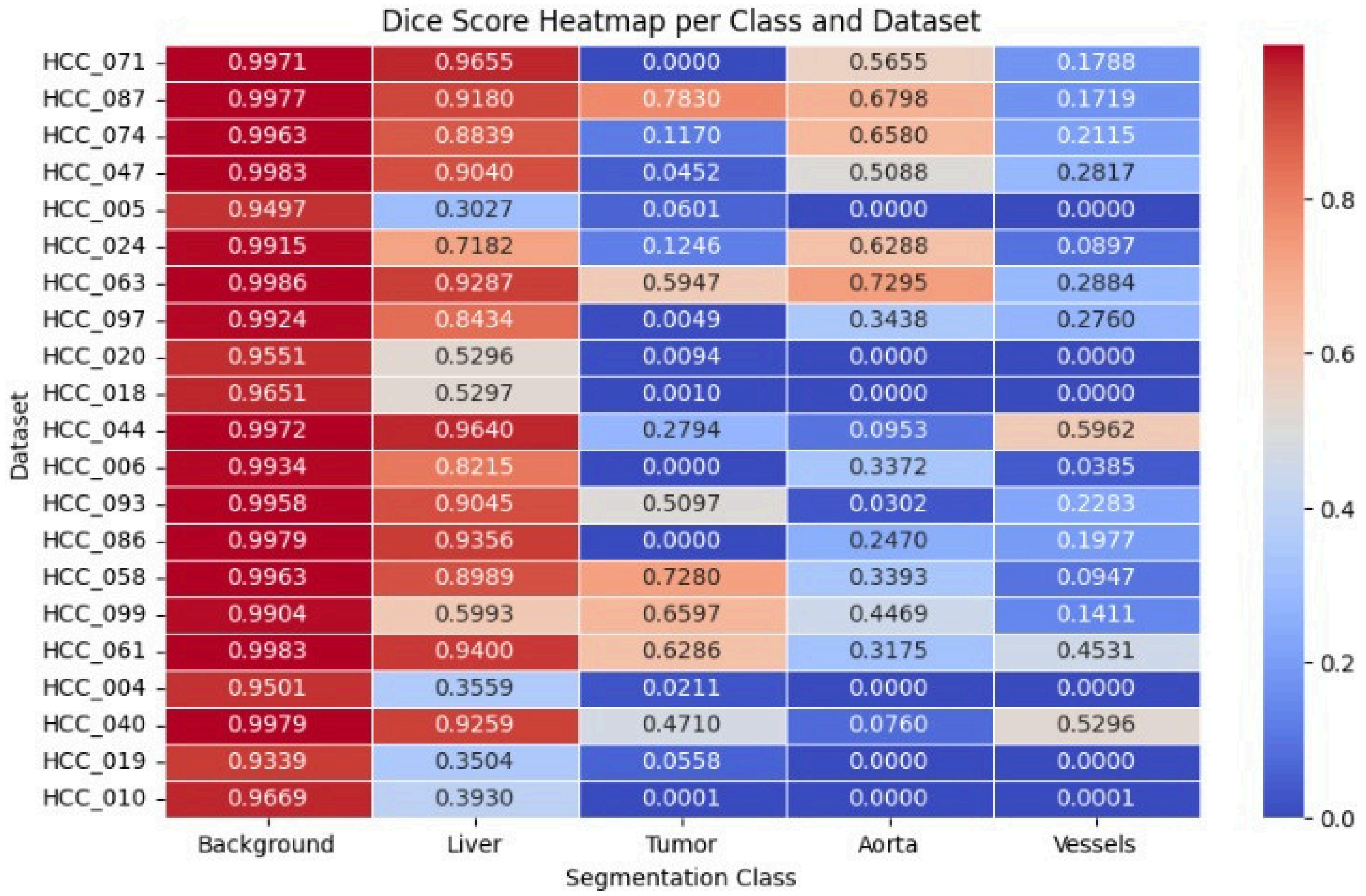
WHEN THIS DICE SCORE IS USED --> (2.0 * intersection + 1e-6) / (union + 1e-6)

[24]:

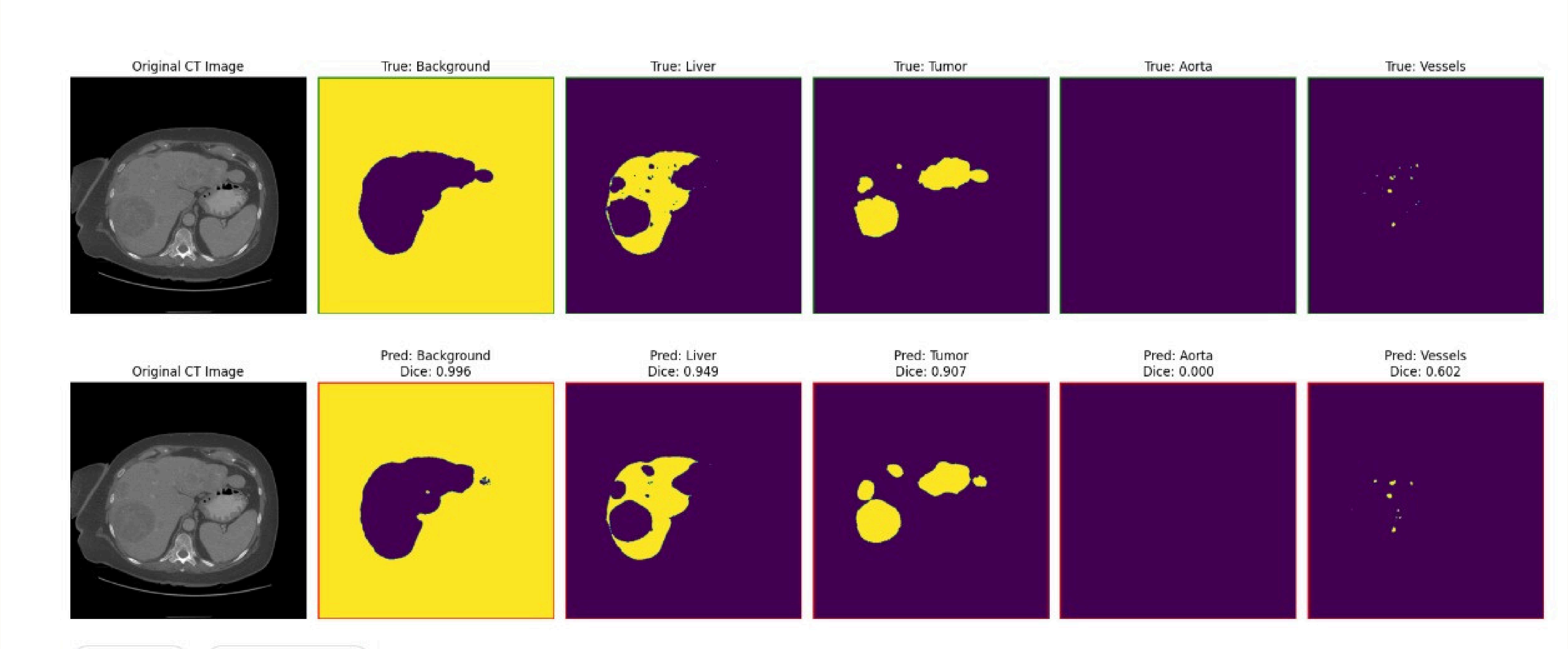
```
# Evaluate Dice coefficients
train_dice_scores = evaluate_dice(unet_model, train_dataset)
print("Train Dice Scores per Class:", train_dice_scores)

test_dice_scores = evaluate_dice(unet_model, test_dataset)
print("Val Dice Scores per Class:", test_dice_scores)

true_test_dice_scores = evaluate_dice(unet_model, true_test_dataset)
print("Test Dice Scores per Class:", true_test_dice_scores)
```

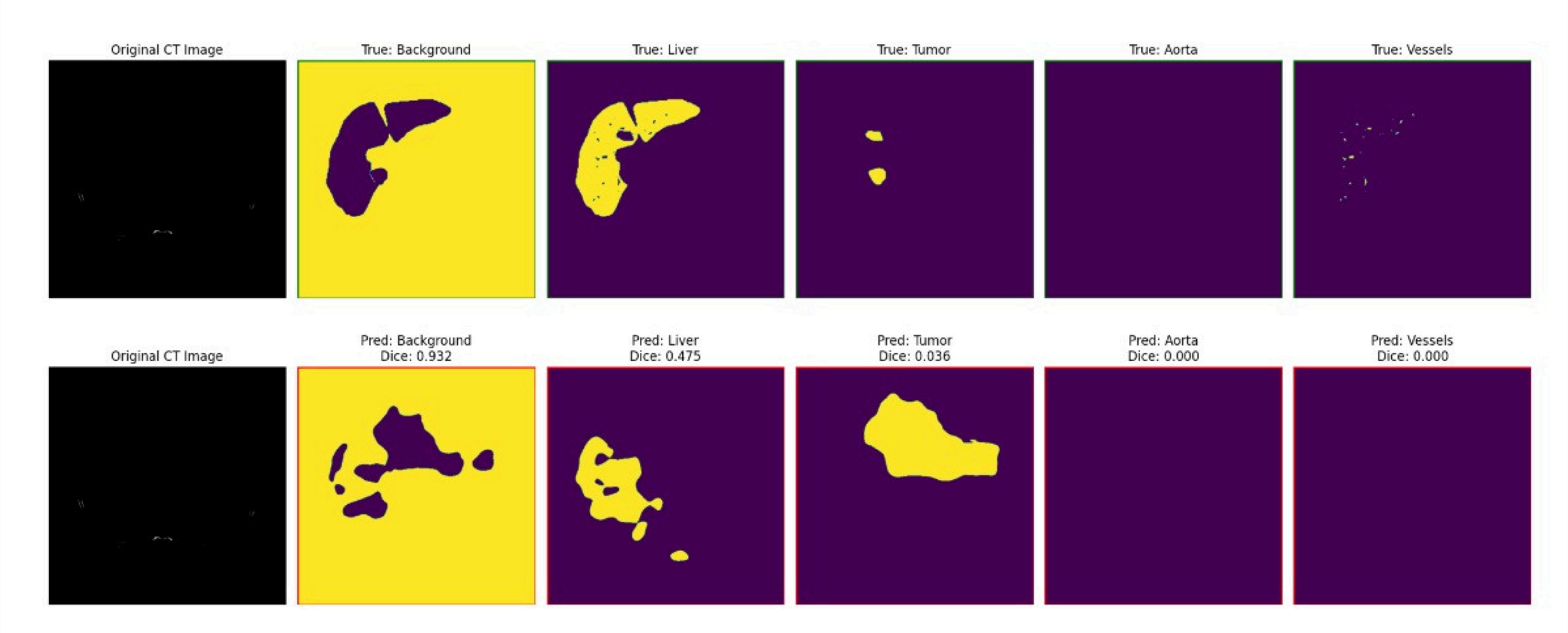
```
Train Dice Scores per Class: {0: 0.9991588252694279, 1: 0.9499322841355947, 2: 0.9745335123233019, 3: 0.99
05299934914156, 4: 0.5710510358611863}
Val Dice Scores per Class: {0: 0.9977042336570983, 1: 0.8827852081031253, 2: 0.8858079372253008, 3: 0.8891
878520708285, 4: 0.5118695718187586}
Test Dice Scores per Class: {0: 0.9878714833295706, 1: 0.7733219581647589, 2: 0.6672582761384052, 3: 0.732
2572915187595, 4: 0.40524012014103117}
```

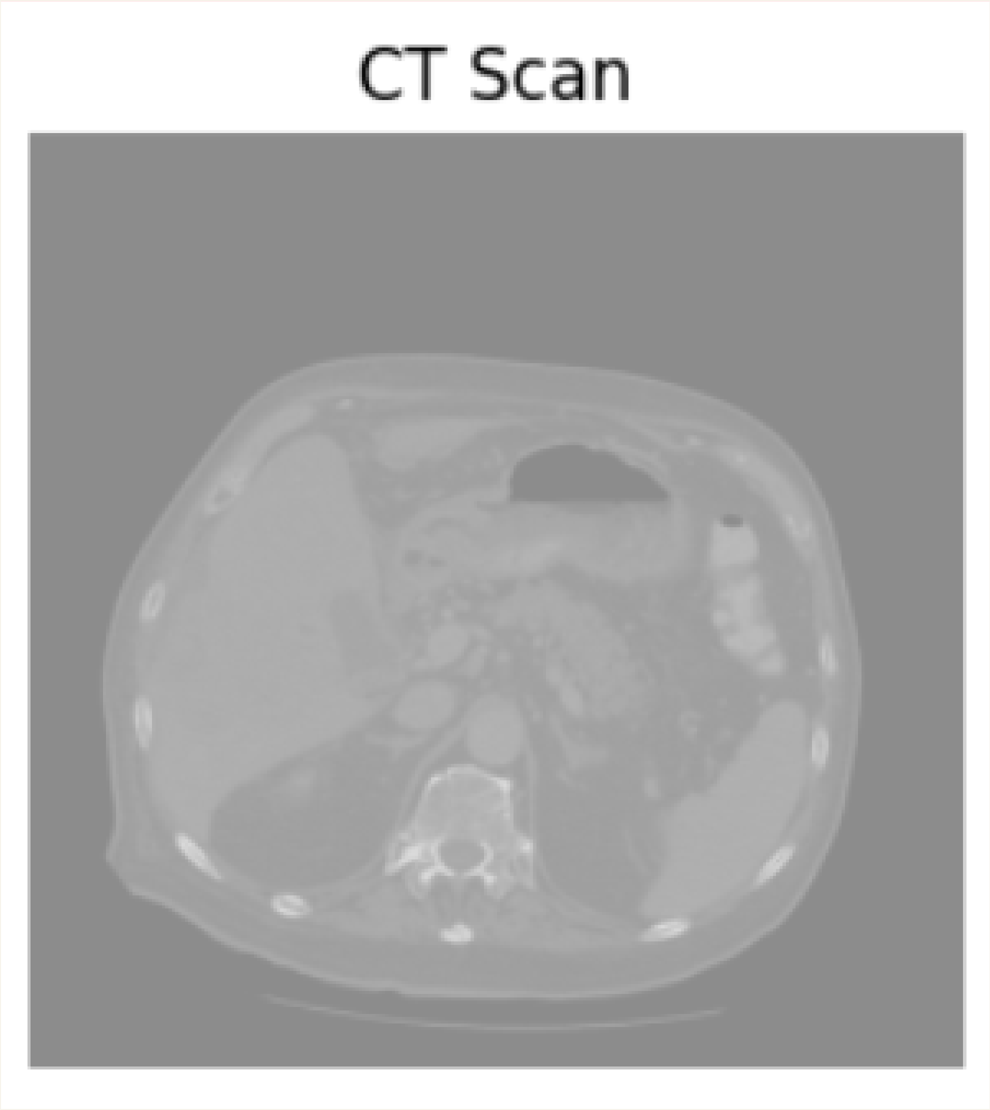
TEST DATA HEAT MAP

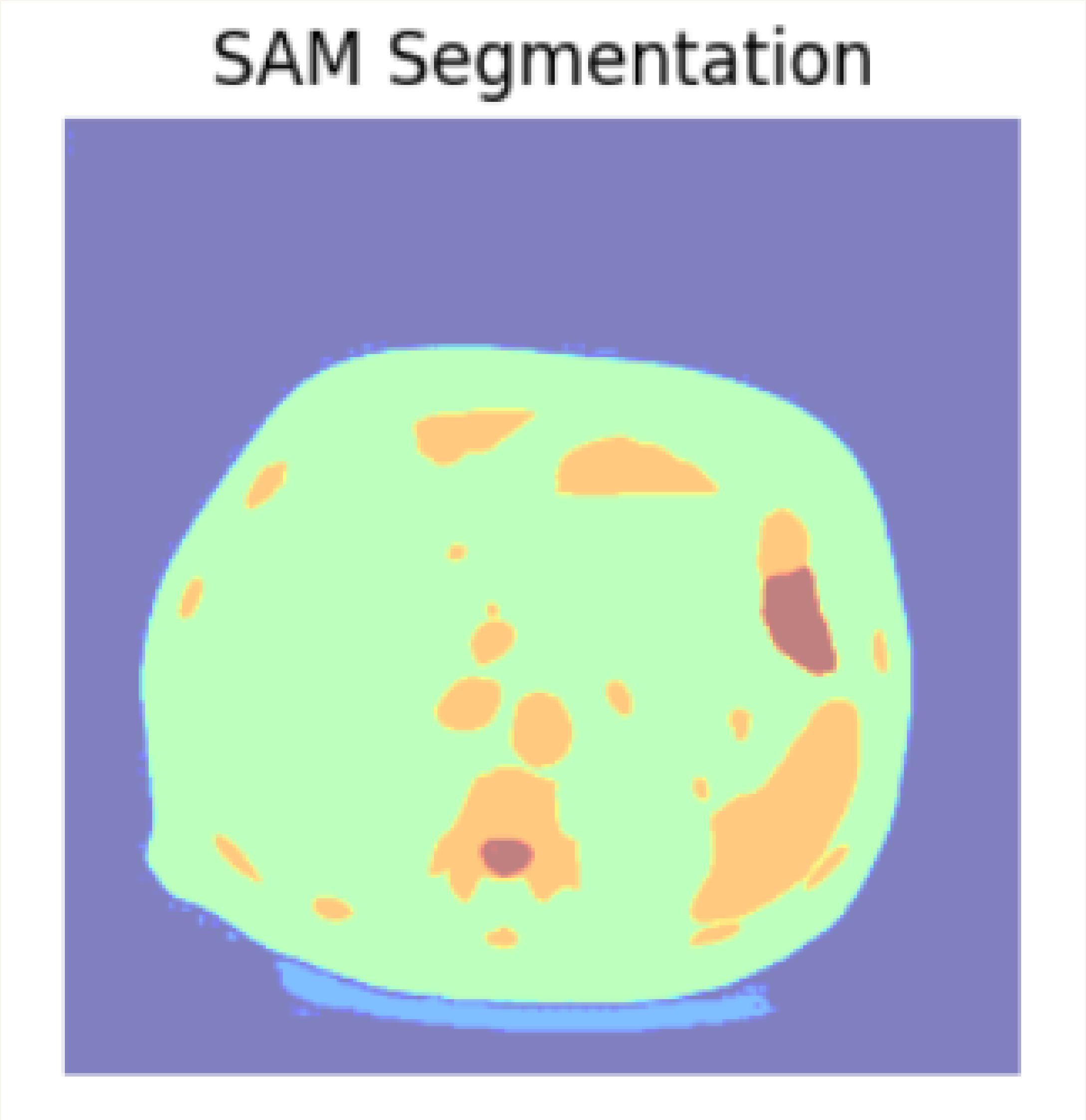


TEST SAMPLES - WINDOWED






CT Scan


SAM Segmentation